

A Graph coloring

PROMPT partial colors 2??1?2 · fill vertices 2, 3, 5

ANSWER 1

113

complete valid coloring

key 211132

ANSWER 2

131

complete valid coloring

key 213112

✓ both correct key 1 ≠ key 2

B Countdown

PROMPT tiles {3, 6, 9} · target 18 · use each tile once

ANSWER 1

$(6 \times 9) / 3$

executes to 18 ✓

AST `div(mul(6,9),3)`

ANSWER 2

$6 + 3 + 9$

executes to 18 ✓

AST `add(9,add(3,6))`

✓ both correct key 1 ≠ key 2

C Python factors

PROMPT lambda n: EXPR · proper factor for 18, 82, 91, 93

ANSWER 1

lambda n: 2 if n%2==0 else
(7 if n%7==0 else 3)

external worker accepts ✓

key [2, 2, 7, 3]

ANSWER 2

lambda n: 3 if n%3==0 else
(13 if n%13==0 else 41)

external worker accepts ✓

key [3, 41, 13, 3]

✓ both correct key 1 ≠ key 2

D MathIR

PROMPT $x/2 - 9 = -1$ · C: subtract -9 · F: $\times 2$ · E: subtract -18

ANSWER 1

C;F

executes to $x = 16$ ✓

key $x/2 = 8 \rightarrow x = 16$

ANSWER 2

F;E

executes to $x = 16$ ✓

key $x - 18 = -2 \rightarrow x = 16$

✓ both correct key 1 ≠ key 2